



SAS821S Capstone Project Charter 2026

T21 Maritime Port and Cargo-Terminal Cyber-Physical Security Analytics

1. Group and project identification

Field	Group response
Topic ID and approved topic	T21 Maritime Port and Cargo-Terminal Cyber-Physical Security Analytics
Working project title	Maritime Port Security Analytics
Member A	215103238 Mwandingi Marshall Data and Modelling Lead
Member B	216086728 Kompeli Ezira Security Engineering and Intelligence Lead
Repository/location	
Date submitted	10 August 2026

2. Problem and decision statement

A modern cargo terminal relies heavily on tight integration of information and operational technologies for safe transfer of containers, recording of cargo information and controlling of yard, gate, vendor and crane operation. In the T21 scenario, the terminal is witnessing abnormal vendor remote access, surprising yard-system commands and intermittent crane control warnings signals. These signals initiate a cyber physical security concern because a compromised vendor account, misconfigured remote session, malicious command series or infected endpoint could affect cargo integrity, operational continuity and physical safety. The problem is not limited to a single log source: the terminal must correlate identity, network, OT/SCADA, cargo and maintenance evidence before deciding whether the activity is benign maintenance, equipment fault, operator error or a cyber-enable attack



The key stakeholders are the terminal security operations team, OT engineers, crane and yard operations supervisors, vendor-management personnel, safety officers, logistics managers and executive decision makers responsible for continuity and risk. If the events are mishandled, the terminal may either overreact and disrupt legitimate operations or underreact and allow unsafe commands, cargo manipulation or service degradation to continue. Therefore, the project will design a defensible security analytics decision support solution that converts multi source evidence into prioritized operational intelligence.

The prototype must support one practical decision: determine whether anomalous vendor remote activity and yard/crane events require escalation, containment and continuity controls, and identify which assets, vendors, cargo processes and safety functions should be prioritized. The solution will use data engineering, baseline behaviors analysis, supervised and unsupervised modelling, event timeline rebuilding, intelligence enhancement, simulation, text mining and predictive risk scoring. The intended outcome is not seamless attribution; it is a transparent, reproducible and safety aware analytical workflow that supports analysts justify whether to separate a vendor session, increase monitoring, block suspicious command patterns, notify operations leadership or continue monitoring under controlled situations.

3. Scope and exclusions

In scope	Out of scope
Synthetic data representing OT/SCADA telemetry, vendor VPN/IAM access, network flows, cargo/manifest events, physical access records, endpoint alerts and maintenance notes. Analysis includes IT/OT correlation, vendor behaviour baselines, command and alarm anomaly detection, incident timeline reconstruction, text mining, predictive risk scoring, simulation of security-control scenarios and analyst-facing decision outputs.	No testing against live ports, vendors, cranes, SCADA systems or third-party networks. No malware execution, credential harvesting, intrusive scanning, real personal, customer data, real production cargo manifests, live safety system control, or claims of legal compliance certification. The project will not attempt to attribute activity to real threat actors beyond high level intelligence mapping.



4. Objectives and security analytics questions

Objective/question	Measure of completion
Objective 1	Create a governed multi-source dataset of at least three security sources and at least 5,000 combined events, including one identity/access source and one OT/network/application source.
Objective 2	Implement supervised and unsupervised/behavioral analytics that produce interpretable alerts with documented evaluation metrics and thresholds.
Question 1	Which vendor sessions, accounts, devices or time windows deviate from normal access behaviour? Evidence: VPN/IAM features, baseline profiles and anomaly scores.
Question 2	Do suspicious access events correlate with yard-system commands, crane-control alarms, cargo/manifest changes or network-flow anomalies? Evidence: joined event timeline and entity relationships.
Question 3	Which control scenario best reduces operational and safety risk while preserving continuity? Evidence: simulation comparing MFA/session restrictions, network segmentation and enhanced monitoring.

5. Learning Sessions 1-10 coverage matrix

Session	Course area	Planned project evidence
1	Foundations of security analytics	Problem framing, analytics lifecycle, data classification, descriptive/diagnostic/predictive/prescriptive mapping and decision statement.
2	Analytical software and tools	Python notebooks/scripts for ingestion, cleaning, joining, visualization and reproducible outputs.



3	Machine-learning fundamentals	Feature design, train/test split, supervised classification, clustering/anomaly method and evaluation metrics.
4	Machine learning in cybersecurity	Detector for suspicious vendor access, command anomalies or cyber-physical incident risk.
5	Analytics and incident response	Correlated incident timeline with affected vendors, devices, commands and response recommendations.
6	Security intelligence	Priority intelligence requirements, enrichment of indicators and operational/executive intelligence outputs.
7	Access analytics	Vendor/user/device baselines, unusual time/location/session duration and privilege-use anomalies.
8	Simulations and security processes	Comparison of at least three control scenarios and their operational/safety trade-offs.
9	Security and text mining	NLP preprocessing of maintenance notes or incident narratives with entity/indicator extraction or topic/classification output.
10	Advanced security intelligence	Predictive risk score, robustness/drift/adversarial tests and governance recommendations.

6. Proposed data sources

Source	Format/volume	Key fields	Provenance/access	Security/ethics controls
Vendor VPN/IAM logs	CSV; about 1,500-2,000 events	timestamp, vendor_id, account_id, source_ip, device_id, auth_result, MFA_status, session_duration, role	Synthetic or approved training data generated to match scenario	No real credentials or IPs; account pseudonymization; documented generation rules.



OT/SCADA and crane-control telemetry	CSV; about 1,500-2,000 events	timestamp, asset_id, command_type, alarm_code, process_state, operator_id, severity	Synthetic OT telemetry based on public schema assumptions	No live OT endpoints; safety-related labels are simulated and non-operational.
Network flow and endpoint/security alerts	CSV/JSON; about 1,500-2,000 events	timestamp, src/dst, ports, protocol, bytes, direction, process_name, alert_type	Public/synthetic flow and alert data	No malware samples; indicators are benign placeholders; limit sensitive host details.
Cargo/manifest and physical access context	CSV; about 800-1,200 events	container_id, yard_zone, gate_event, badge_id, cargo_status, manifest_change	Synthetic contextual records	No real cargo/customer data; pseudonymised container and badge identifiers.
Text data source: maintenance notes and incident narratives	CSV/text; at least 200 records	note_id, timestamp, asset_id, vendor_id, free_text, category, priority	Synthetic notes plus public threat-report summaries where permitted	Remove personal data; cite public text; do not include exploit instructions.



7. Proposed analytical methods

Component	Proposed method/tool	Expected output
Baseline/EDA	Python, pandas, matplotlib/seaborn; descriptive statistics and event-rate baselines	Normal access windows, asset alarm baselines, event volumes, top entities and two or more security visualizations.
Supervised ML	Random Forest or Logistic Regression on labelled suspicious/benign events	Confusion matrix, precision, recall, F1-score and ranked high-risk events.
Unsupervised or UBA	Isolation Forest, One-Class SVM or clustering for vendor/device behaviour	Anomaly scores, threshold selection and analyst interpretation.
Incident investigation	Entity-time correlation and timeline reconstruction	Incident timeline linking vendor access, network flows, commands, alarms and cargo context.
Security intelligence	PIRs, MITRE ATT&CK for ICS mapping and IOC/entity enrichment	Operational intelligence note and executive risk summary.
Simulation	What-if or Monte Carlo comparison of three controls	Scenario comparison for MFA/session restrictions, segmentation and enhanced monitoring.
Text mining/NLP	Tokenization, cleaning, TF-IDF/entity extraction/topic or classification model	Extracted asset IDs, alarm terms, vendor mentions, topics/categories and validation summary.
Predictive/adversarial analysis	Risk scoring, drift checks and three manipulation tests	Early-warning score and robustness notes for evasion, missing data and label noise.

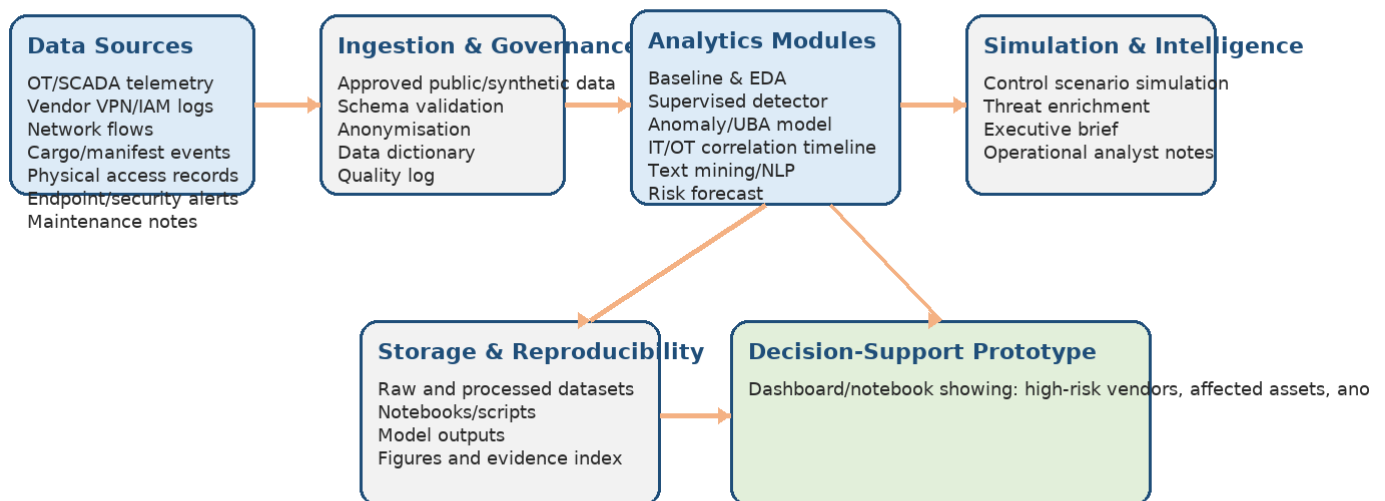


Prototype/dashboard	Jupyter notebook, Streamlit dashboard or Power BI-style mock-up	Analyst view of alerts, timelines, risk scores, evidence and recommended decisions.
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8. High-level architecture

The architecture below demonstrates how approved data sources will be governed, ingested, transformed, analysed and presented as analyst decision support.

T21 Security Analytics Decision-Support Architecture



Decision supported: prioritise whether vendor remote activity and yard/crane anomalies require containment, operational continuity actions, and safety

9. Roles and RACI

Workstream	Member A	Member B	Review/approval evidence
Data acquisition and quality	A/R	C	Data inventory, data dictionary, cleaning log and notebook commits.



ML and anomaly modelling	A/R	C/R for security validation	Experiment log, metrics, thresholds and peer review notes.
Investigation and access analytics	C	A/R	Incident timeline, access-analysis outputs and validation comments.
Simulation and intelligence	R for analytics support	A/R	Simulation assumptions, intelligence brief and scenario comparison.
Prototype and testing	R	R	Prototype screenshots, test cases and issue log.
Documentation and presentation	R	R	Draft history, signed contribution declaration and presentation outline.

10. Work plan

Task	Owner	Start	Finish	Dependency	Output
Finalize charter, topic scope and data plan	Both	04 Aug 2026	10 Aug 2026	Template and assessment brief	Approved Milestone 1 charter PDF.
Generate/acquire datasets and data dictionary	Member A	11 Aug 2026	22 Aug 2026	Charter approval	Raw/processed data, provenance record and dictionary.
Build baseline, supervised and anomaly models	Member A	23 Aug 2026	05 Sep 2026	Clean data	EDA, models, metrics and experiment log.
Develop investigation, intelligence, NLP and simulation modules	Member B	23 Aug 2026	08 Sep 2026	Clean data and hypotheses	Timeline, intelligence outputs, NLP output and simulation.



Integrate dashboard/prototype and test cases	Both	09 Sep 2026	18 Sep 2026	Analytical outputs	Working prototype, screenshots and test evidence.
Prepare implementation bundle and final review	Both	19 Sep 2026	20 Sep 2026	Prototype complete	Milestone 2 implementation package.

11. Risk and ethics register

Risk	Likelihood	Impact	Mitigation	Owner
Data unavailable or unsuitable	M	H	Use approved synthetic data plan; document assumptions and seek Facilitator approval for smaller or specialist datasets.	A
Privacy or sensitive-data exposure	L	H	Use pseudonymized synthetic only; remove personal, credential and production records.	Both
Model bias or misleading results	M	M	Validate with confusion matrix, threshold analysis, false-positive review and limitations section.	A
Scope/time overrun	M	H	Prioritize mandatory components, maintain issue log and use minimum viable prototype approach.	Both
Technical integration failure	M	M	Use reproducible notebooks, small modular scripts, version control and fallback static outputs.	B



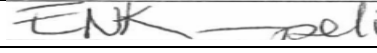
Unequal contribution	L	M	RACI allocation, meeting notes, commit history and signed contribution statement.	Both
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12. Responsible AI and academic integrity statement

The team will make use of AI and coding assistants for brainstorming, editing, syntax check, and quality assurance purposes. All text produced, code produced, output and external references will be edited, checked, and verified by the team members before submission. Open-source data sets, reused code and external references will be cited. Guidelines for synthetic data creation will be provided. The project will not be making use of any unauthorized production data, personal data, genuine credentials, API key data, malware sample data and intrusive testing against live entities.

13. Group agreement and declaration

- We confirm that this group consists of exactly two students.
- We accept the role of allocation and will retain evidence of our contributions.
- We understand that both members must explain the complete solution during the presentation and Q&A.
- We will use only authorized data and will not test live third-party systems.
- We will disclose permitted AI assistance, reused code and external data sources.

Member	Signature	Date
Mwandingi Marshall		10 August 2026
Kompeli Ezira		10 August 2026

14. Lecturer approval

Decision	Comments/conditions	Lecturer/date
Approved / Revise / Not approved		

